Revant Teotia

rt2741@nyu.edu | revantteotia.github.io | linkedin.com/in/revantteotia

Education

NYU Courant Institute of Mathematical Sciences — New York, NY
Ph.D. in Computer Science — *Sep 2023 -*

Columbia University — New York, NY
M.S. in Computer Science — *Sep 2021 - Dec 2022*

Indian Institute Of Technology Kanpur — Kanpur, India
B.Tech. in Computer Science and Engineering — *Jul 2013 - May 2017*

Publications

- MJEPA: A Simple and Scalable Joint-Embedding Predictive Architecture for Audio-Visual Learning, [paper] [code]
  Revant Teotia, Adrien Bardes, Michael Rabbat, Sumit Chopra, Matthew J. Muckley, Nicolas Ballas,
  ECCV 2026

- DIMCIM: A Quantitative Evaluation Framework for Default-mode Diversity and Generalization in Text-to-Image Generative Models, [paper] [code]
  Revant Teotia, Candace Ross, Karen Ullrich, Sumit Chopra, Adriana Romero-Soriano, Melissa Hall, Matthew J. Muckley,
  ICCV 2025

- Codesign of Tensors Encoding And Transcoding: A Building Block For Decentralized AI, [paper]
  Revant Teotia, Muhammad Haseeb,
  Workshop on Networks for AI Computing, ACM SIGCOMM 2025

- Affective Faces for Goal-Driven Dyadic Communication, [arXiv] [project page]
  Scott Geng, **Revant Teotia**, Purva Tendulkar, Sachit Menon, Carl Vondrick,
  In submission

- Doubly Right Object Recognition: A Why Prompt for Visual Rationales, [paper]
  Chengzhi Mao, **Revant Teotia**, Amrutha Varshini Sundar, Sachit Menon, Junfeng Yang, Xin Wang, Carl Vondrick,
  CVPR 2023

- COFAR: Commonsense and Factual Reasoning in Image Search, [paper][project page]
  Prajwal Gatti, Abhirama Penamakuri, **Revant Teotia**, Anand Mishra, Shubhashis Sengupta, Roshni Ramnani,
  AACL-IJCNLP 2022

- Doubly Right Object Recognition, [paper]
  Revant Teotia, Chengzhi Mao, Carl Vondrick,
  ICML 2022 Workshop on Spurious Correlations, Invariance, and Stability (**Spotlight talk, among top-5 papers**)

- Finding Spuriously Correlated Visual Attributes, [paper]
  Revant Teotia, Chengzhi Mao, Carl Vondrick,
  ICML 2022 Workshop on Spurious Correlations, Invariance, and Stability

- Few-shot Visual Relationship Co-localization, [paper][project page][code]
  Revant Teotia, Vaibhav Mishra, Mayank Maheshwari, Anand Mishra,
  ICCV 2021

- Realtime Indoor Workout Analysis Using Machine Learning & Computer Vision, [paper]
  Amit Nagarkoti, **Revant Teotia**, Amith K. Mahale, and Pankaj K. Das,
  IEEE EMBC 2019

Experience

FAIR Labs, Meta AI — New York, NY
Visiting Researcher — *Sep 2024 - Aug 2026*

- Worked on multimodal self-supervised representation learning methods.
- Worked on image generation model evaluations.

Langone Radiology Department, New York University — New York, NY
PhD student researcher in Sumit Chopra's Lab — *Sep 2023 - Present*

- Working on developing machine learning techniques for radiology applications.
- Developing self-supervised learning methods for breast MRI for early and improved breast cancer detection.
- Developing multimodal representation learning methods for cariology using EKG/Echo/other heart monitoring data.

Computer Science Department, Columbia University — New York, NY
Research Assistant in Carl Vondrick's group — *Sep 2021 - Aug 2023*

- Explainable models: making object recognition models explainable using object descriptions. Training visual prompt for large vision-language model CLIP to give right reasons for predictions.
- Social AI: Understanding human emotions, cultures and interactions from unlabeled videos. Using implicit social knowledge in large language model GPT-3 with vision-language model CLIP to model human interactions.

Indian Institute of Technology Jodhpur — Jodhpur, India
Research Assistant in Anand Mishra's Vision, Language, and Learning Group (VL2G) — *Jul 2020 - Aug 2021*

- Analyzed novel problem of Visual Relationship Co-localization and invented a meta-learning based optimization framework to solve it in a few-shot manner. Published in ICCV 2021. [paper][project page][code]
- Worked on incorporating commonsense and factual reasoning in image search using external knowledge sources. Published in AACL-IJCNLP 2022. [paper][project page]

Samsung Research Institute — Bangalore, India
Senior Software Engineer, Samsung Health Team — *Jul 2017 - Jul 2019*

- Designed and developed the C++ modules of BLE/NFC communication between Samsung smartwatch and gym fitness equipments. Announced at CES2020 and enjoyed by millions of Samsung Galaxy series 2020 smartwatch users. [news]
- Created a home workout video analysis system using CNN-based human pose-estimation, Optical Flow and Dynamic Time Warping algorithm. Published in IEEE EMBC 2019. [paper]